

ARNOB RIZWAN AHMAD

AI / LLM Engineer • Agentic Systems • RAG & Evaluation • Python Backend • Cloud

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PROFILE

AI/LLM engineer and Software Engineering graduate of Universiti Teknologi Malaysia (July 2026), moving into AI from production full-stack and ERP delivery. Build LLM systems with that same operational discipline — agentic workflows, RAG over FAISS and pgvector, and evaluation harnesses with blinded LLM-judge scoring and execution guardrails, on the FastAPI, PostgreSQL and Kubernetes infrastructure they run on. Certified AMD ROCm Associate. Seeking AI/ML engineering roles where evaluation and reliability matter as much as model choice.

EDUCATION

Universiti Teknologi Malaysia (UTM) — Johor Bahru

Oct 2022 – July 2026

Bachelor of Software Engineering (Computer Science) — Graduated July 2026

PROFESSIONAL EXPERIENCE

Agentic AI Engineer — BLOOMTECH, Inc. — Remote (Part-time)

Aug 2026 – Present

Agentic AI systems and LLM-driven tooling

- Build agentic AI workflows in Python — multi-step LLM planning, tool and API integration, and structured output handling.
- Design prompt and evaluation loops that measure and improve agent reliability across task categories.

Enterprise Resource Planning Developer — Hirdaramani Bangladesh — Hybrid (Full-time)

Jul 2026 – Present

ERP systems for a multinational apparel manufacturing group

- Build and maintain ERP modules supporting production, inventory and order-management workflows across factory operations.
- Write and tune SQL against large operational datasets; build reports and data integrations between ERP and internal systems.
- Work with operations and IT stakeholders to translate process requirements into shipped features.

Application Developer (Freelance) — Moien Delivery — Luxembourg (Remote)

Mar 2026 – Jul 2026

Hybrid logistics platform serving customers, restaurants and drivers — live on Google Play & Apple App Store

- Built the Angular web front-end and a cross-platform React Native mobile app on a Nuxt.js backend, delivering features for three distinct user roles.
- Implemented real-time order tracking, driver dispatch and delivery workflow automation over REST APIs and WebSocket event streams.
- Owned features from spec through QA to production deployment on weekly release cycles with international stakeholders.

Machine Learning Engineer (Contract via Mercor) — AI News HUD — USA (Remote)

Oct 2025 – Mar 2026

US-based AI-powered news intelligence platform

- Built and evaluated production PyTorch models for real-time news classification, summarization and personalized recommendations.
- Designed evaluation pipelines and ran iterative experiments on live traffic, raising news-classification accuracy from **70%** to **78.9%** and cutting inference latency **20%**.
- Took ML components from prototype to deployed feature in the platform's core intelligence layer.

Freelance & Contract Engineering

Sep 2024 – Jul 2026

Concurrent client and programme work delivered alongside study

- **Framic** (remote, Aug 2025 – Jul 2026, ~10 hrs/week) — founding backend engineer at an AI storage startup; designed backend services from scratch, owned architecture and core API design, and led AI search and real-time file indexing.
- **MOMS Mobile Oil Change** (USA, Apr – Jul 2026) — production Next.js site, live at momssoilchange.com: rebuilt navigation and three product pages, led an SEO overhaul, and ran a site-wide security and dependency audit.
- **Headstarter** (USA, Sep 2024 – Mar 2025) — competitive 6-month software engineering residency; shipped ML and full-stack projects end to end, and cut image size and build time on a Docker-based GitHub Actions pipeline.

SELECTED PROJECTS

Al-Ibadah — Bangla Islamic Reference Platform (alibadahfoundation.com)

Sep 2026

- Shipped a Bangla-first Islamic reference site, live at alibadahfoundation.com: the full Quran with five Bangla translations, three tafsirs and ten reciters, 36,432 hadith across 8 collections, duas, and prayer times and qibla for all 64 districts.
- Built an in-browser question-answering assistant over a hand-written, source-cited corpus — BM25 plus character-trigram retrieval with a confidence threshold that refuses rather than fabricates, needing no model or API key at runtime.
- React 18, Vite and TypeScript exported to a fully static build with all 141 routes prerendered for SEO, deployed on GitHub Pages with a custom domain and HTTPS.

School Care App — 40-School Deployment — Sylhet, Bangladesh

Mar 2026

- Delivered a full-stack school management system adopted by 40 schools across Sylhet; in closed testing ahead of Google Play launch.
- Built a React admin dashboard and a React Native app for parents and teachers, with Cloud Functions fanning out role-targeted FCM push notifications by school, role, and class.

ShopLocalr — Local Commerce Mobile App — Malaysia

Feb 2026

- Shipped a Flutter local-business discovery app with customer, business, and admin roles — QR deal redemption with checkout PINs, Gemini-backed recommendations, and offline-first Hive caching on a Firebase backend.

Autonomous ML Agent — LLM-Planned AutoML Pipeline (GitHub)

- Built an AutoML system where an LLM planner proposes modeling strategies under a time budget, Optuna tunes XGBoost/LightGBM/scikit-learn models, ensembles are built, and SHAP explains the winner.
- Shipped with model cards, a FastAPI inference service, an automated test suite, and CI/CD publishing to Docker Hub.

Self-Improving Prompt Optimization API — LLM Evals Infrastructure (GitHub)

- Built CI/CD for prompts: a semver-versioned prompt registry where each improvement cycle runs failure analysis, generates candidates, A/B-evaluates them with a blinded LLM judge, and auto-promotes winners per YAML rules.
- Multi-dimensional scoring across OpenAI and Anthropic providers with FastAPI, Postgres, and Redis.

CrisisCohort — Multi-Agent AI Climate Resilience Platform (GitHub)

- Architected a climate-resilience platform for smallholder farmers with 10+ specialized AWS Bedrock agents (water optimization, climate migration, grant matching, disease diagnostics) deployed on AWS Lambda.
- Designed an AWS architecture spanning Lambda, Bedrock, DynamoDB, CloudFront, API Gateway, and Cognito, plus an on-chain water-rights trading contract in Solidity.

Terminal Coding Agent — Sandboxed CLI Agent (GitHub)

- Built a pip-installable CLI coding agent that plans edits via OpenAI/Anthropic, applies them through a diff manager, and executes inside E2B Firecracker microVMs.
- Added guardrails that scan for leaked credentials (AWS, GitHub, Slack) before any edit is applied, covered by an automated test suite.

TECHNICAL SKILLS

ML / AI: PyTorch, scikit-learn, XGBoost, LightGBM, Optuna, SHAP; LLM agents, RAG (FAISS, pgvector), LLM evaluation & blinded judges, prompt engineering, AWS Bedrock, Gemini

GPU / ROCm: AMD ROCm stack and HIP programming model (ROCm Certified Associate, 2026)

Languages: Python, SQL, TypeScript, JavaScript, C#, Java, Dart

Backend: FastAPI, Django + DRF, Node.js, Express, NestJS, ASP.NET Core, Celery, REST APIs, WebSockets

Frontend & Mobile: React, Next.js, Angular, React Native, Flutter

Cloud & Data: AWS, Microsoft Azure, GCP, Docker, Kubernetes, Linux, CI/CD (GitHub Actions, Jenkins); PostgreSQL, MySQL, SQL Server, MongoDB, Redis, ERP modules and SQL tuning on large operational datasets

CERTIFICATIONS

- **AMD — ROCm Certified Associate** (Aug 2026)
- **AWS** — Building Language Models on AWS; Customizing & Evaluating LLMs Using Amazon SageMaker JumpStart; Amazon SageMaker JumpStart Foundations; Foundations of Prompt Engineering (Mar 2025)
- **IBM** — Full Stack Software Developer Professional Certificate (Aug 2024)
- **Microsoft Azure AI** — Azure OpenAI Service, Natural Language Processing, Document Intelligence (Jun 2024)